



PREMIER UNIVERSITY CHATTOGRAM

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LAB REPORT

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1. Introduction

Air pollution is one of the major environmental and public health concerns in many parts of the world. Air quality monitoring systems collect large amounts of data about different pollutants, monitoring locations, measurement dates, and pollutant concentrations. As the number of monitoring stations and the duration of data collection increase, the volume of air-quality data becomes very large and difficult to process efficiently using traditional data-processing methods. This project, “Air Quality Big Data Analysis using Hadoop,” focuses on processing and analyzing a large air-quality dataset using Big Data technologies. The dataset contains approximately 1.69 million records collected over 19 years. The large volume of records provides an appropriate environment for demonstrating distributed data processing using the Hadoop ecosystem.

The project uses Apache Hadoop as the main Big Data processing framework. Hadoop Distributed File System (HDFS) is used to store the large dataset in a distributed manner. HDFS allows large files to be divided into blocks and stored across the Hadoop environment, making it suitable for handling large datasets. The dataset is uploaded to HDFS before performing the analysis.

MapReduce is used to perform several analyses on the air-quality data. Three MapReduce jobs are implemented in this project. The first job calculates the average pollutant concentration, the second identifies the highest pollution location, and the third analyzes the pollution trend. These jobs demonstrate how large datasets can be processed using distributed computation.

In addition to MapReduce, Apache Pig is used to perform further analysis. Five Pig tasks are implemented: Average Pollutant Analysis, Location Analysis, Pollution Trend Analysis, Top Polluted Locations, and Pollutant Analysis. Pig Latin provides a high-level approach for loading, filtering, grouping, and analyzing large datasets without requiring every operation to be implemented using low-level MapReduce programming.

2. Project Objectives

No.	Objective
1	Store the large air-quality dataset in HDFS.
2	Process approximately 1.69 million records using Hadoop.
3	Calculate average pollutant concentrations using MapReduce.
4	Identify highest pollution locations from the dataset.
5	Analyze pollution trends over time.
6	Perform additional analysis using Apache Pig.
7	Verify the successful execution of Hadoop, MapReduce, and Pig jobs.
8	Demonstrate the use of Big Data technologies for environmental data analysis.

3. Problem Statement

Air-quality monitoring produces a large amount of data containing information about different pollutants, monitoring locations, dates, and measured pollutant values. When this data is collected continuously over many years, the resulting dataset becomes large and difficult to analyze efficiently using traditional data-processing techniques.

The main problem addressed by this project is how to store, process, and analyze a large air-quality dataset efficiently using Big Data technologies. Traditional single-machine processing can become time-consuming when working with millions of records, especially when multiple types of analysis need to be performed on the same dataset.

The project specifically aims to solve the following problems:

- How to store a large air-quality dataset using Hadoop Distributed File System (HDFS).
- How to calculate average pollutant concentrations from millions of records.
- How to identify locations with the highest recorded pollution values.
- How to analyze pollution trends over time.
- How to perform additional air-quality analysis using Apache Pig.
- How to verify that Hadoop, HDFS, MapReduce, and Pig are correctly configured and successfully process the dataset.

4. Dataset Name and Description

Dataset Link: <https://www.kaggle.com/datasets/epa/epa-historical-air-quality?>

Item	Description
Dataset Name	Air Quality Dataset
Source	U.S. Environmental Protection Agency (EPA) AirData
Time Period	19 years
Number of Records	Approximately 1.69 million
Dataset Size	Approximately 879 MB
File Format	CSV
Main Data	Pollutant measurements, locations, dates, and pollutant information
Storage	Hadoop Distributed File System (HDFS)
Processing	MapReduce and Apache Pig

Item	Description
Tools	
Purpose	Analysis of pollutant concentration, pollution locations, and pollution trends

5. Step 1 — Prepare and Verify the Environment

Before processing the air quality dataset, the required Big Data environment must be prepared and verified. The project uses Hadoop HDFS, Apache Pig, and Hadoop MapReduce for storing and analyzing the large dataset.

5.1 Required Software

Software	Purpose
Java JDK	Required to run Hadoop
Hadoop	HDFS storage and MapReduce processing
Apache Pig	Data preprocessing and analysis
Python	Used for MapReduce mapper and reducer programs
WSL/Linux	Provides a suitable environment for running Hadoop tools

5.2 Java version verification:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ java -version
openjdk version "1.8.0_502"
OpenJDK Runtime Environment (build 1.8.0_502-8u502-ga~us1-0ubuntu1~26.04-b07)
OpenJDK 64-Bit Server VM (build 25.502-b07, mixed mode)
```

5.3 Hadoop version verification:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hadoop version
Hadoop 3.2.4
Source code repository Unknown -r 7e5d9983b388e372fe640f21f048f2f2ae6e9eba
Compiled by ubuntu on 2022-07-12T11:58Z
Compiled with protoc 2.5.0
From source with checksum ee031c16fe785bbb35252c749418712
This command was run using /mnt/c/hadoop/share/hadoop/common/hadoop-common-3.2.4.jar
```

5.4 Pig version verification:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ pig -version
Apache Pig version 0.17.0 (r1797386)
compiled Jun 02 2017, 15:41:58
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ _
```

6. Step 2 — Hadoop Service Verification

The Hadoop services were started using the required commands. The `jps` command was then used to verify that the Hadoop processes were running successfully.

6.1 Hadoop services running in jps:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ jps
4529 NodeManager
5033 Jps
4409 ResourceManager
3177 SecondaryNameNode
3019 DataNode
2924 NameNode
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$
```

6.2 Start YARN:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
```

6.3 Hadoop localhost HDFS browser Directory:

Hadoop	Overview	Datanodes	Datanode Volume Failures	Snapshot	Startup Progress	Utilities
Overview 'localhost:9000' (active)						
Started:	Mon Sep 14 20:10:35 +0600 2026					
Version:	3.2.4, r7e5d9983b388e372fe640f21f048f2f2ae6e9eba					
Compiled:	Tue Jul 12 17:58:00 +0600 2022 by ubuntu from branch-3.2.4					
Cluster ID:	CID-3f9f462e-a2b0-4447-8be1-165a514bbd7					
Block Pool ID:	BP-787223300-127.0.1.1-1789029273100					

7. Map Reduce Analysis :

Merge Files : Merges all the CSV Files into one

Code:

```
the last disk(s) of this archive.
note: annual_conc_by_monitor_2011.zip may be a plain executable, not an archive
unzip: cannot find zipfile directory in one of annual_conc_by_monitor_2011.zip or
annual_conc_by_monitor_2011.zip.zip, and cannot find annual_conc_by_monitor_2011.zip.ZIP, period.
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis/dataset/raw$ ls *.csv | wc -l
19
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis/dataset/raw$ ls *.csv | wc -l
19
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis/dataset/raw$ head -n 1 annual_conc_by_monitor_2000.csv > Air_Quality.csv
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis/dataset/raw$ for file in annual_conc_by_monitor_*.csv; do
tail -n +2 "$file" >> Air_Quality.csv
done
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis/dataset/raw$ ls -lh Air_Quality.csv
-rwxrwxrwx 1 aceracer aceracer 879M Sep 14 20:04 Air_Quality.csv
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis/dataset/raw$ wc -l Air_Quality.csv
1691537 Air_Quality.csv
```

Part A · MapReduce (3 jobs)

7.1 MR1 — Average Pollutant Concentration

Execution:

```
aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hadoop jar /mnt/c/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.2.4.jar \
-input /air_quality/input/Air_Quality.csv \
-output /air_quality/output/MR1_AveragePollutant \
-mapper /mnt/d/Air_Quality_Big_Data_Analysis/mapreduce/MR1_AveragePollutant/mapper.py \
-reducer /mnt/d/Air_Quality_Big_Data_Analysis/mapreduce/MR1_AveragePollutant/reducer.py
packageJobJar: [/tmp/hadoop-unjar3966422993827698570/] [] /tmp/streamjob3820461799113433077.jar tmpDir=null
2026-09-14 21:00:45,155 INFO client.RMProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-09-14 21:00:46,087 INFO client.RMProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-09-14 21:00:46,826 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/aceracer/.staging/job_1789397904369_0001
2026-09-14 21:00:47,770 INFO mapred.FileInputFormat: Total input files to process : 1
2026-09-14 21:00:47,959 INFO mapreduce.JobSubmitter: number of splits:7
2026-09-14 21:00:48,168 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789397904369_0001
2026-09-14 21:00:48,176 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-14 21:00:48,788 INFO conf.Configuration: resource-types.xml not found
2026-09-14 21:00:48,791 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-14 21:00:49,052 INFO impl.YarnClientImpl: Submitted application application_1789397904369_0001
2026-09-14 21:00:50,025 INFO mapreduce.Job: The url to track the job: http://DESKTOP-P503RAM.localdomain:8088/proxy/application_1789397904369_0001/
2026-09-14 21:00:50,034 INFO mapreduce.Job: Running job: job_1789397904369_0001
2026-09-14 21:01:12,349 INFO mapreduce.Job: Job job_1789397904369_0001 running in uber mode : false
2026-09-14 21:01:12,351 INFO mapreduce.Job: map 0% reduce 0%
2026-09-14 21:01:46,126 INFO mapreduce.Job: map 80% reduce 0%
2026-09-14 21:02:01,254 INFO mapreduce.Job: map 100% reduce 0%
2026-09-14 21:02:06,291 INFO mapreduce.Job: map 100% reduce 100%
2026-09-14 21:02:07,310 INFO mapreduce.Job: Job job_1789397904369_0001 completed successfully
2026-09-14 21:02:07,726 INFO mapreduce.Job: Counters: 55
File System Counters
FILE: Number of bytes read=50431347
FILE: Number of bytes written=102784107
FILE: Number of read operations=0
FILE: Number of large read operations=0
```

Result:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -ls /air_quality/output/MR1_AveragePollutant
Found 2 items
-rw-r--r-- 1 aceracer supergroup 0 2026-09-14 21:02 /air_quality/output/MR1_AveragePollutant/_SUCCESS
-rw-r--r-- 1 aceracer supergroup 31255 2026-09-14 21:02 /air_quality/output/MR1_AveragePollutant/part-00000
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$
```

Hadoop

Overview

Datanodes

Datanode Volume Failures

Snapshot

Startup Progress

Utilities

Browse Directory

/air_quality/output/MR1_AveragePollutant

Go!



Show 25 entries

Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	☐
☐	-rw-r--r--	aceracer	supergroup	0 B	Sep 14 21:02	1	128 MB	_SUCCESS	☐
☐	-rw-r--r--	aceracer	supergroup	30.52 KB	Sep 14 21:02	1	128 MB	part-00000	☐

Showing 1 to 2 of 2 entries

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Hadoop, 2022.

7.2 MR2 — Highest Pollution Location.

Execution:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hadoop jar /mnt/c/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.2.4.jar \
-input /air_quality/input/Air_Quality.csv \
-output /air_quality/output/MR2_HighestPollutionLocation \
-mapper /mnt/d/Air_Quality_Big_Data_Analysis/mapreduce/MR2_HighestPollutionLocation/mapper.py \
-reducer /mnt/d/Air_Quality_Big_Data_Analysis/mapreduce/MR2_HighestPollutionLocation/reducer.py
packageJobJar: [/tmp/hadoop-unjar2440029489471267439/] [] /tmp/streamjob2525167295214968284.jar tmpDir=null
2026-09-14 21:19:01,428 INFO client.RMProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-09-14 21:19:02,964 INFO client.RMProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-09-14 21:19:03,818 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/aceracer/.staging/job_1789397904369_0002
2026-09-14 21:19:04,059 INFO mapred.FileInputFormat: Total input files to process : 1
2026-09-14 21:19:04,791 INFO mapreduce.JobSubmitter: number of splits:7
2026-09-14 21:19:05,023 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789397904369_0002
2026-09-14 21:19:05,031 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-14 21:19:05,652 INFO conf.Configuration: resource-types.xml not found
2026-09-14 21:19:05,656 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-14 21:19:05,901 INFO impl.YarnClientImpl: Submitted application application_1789397904369_0002
2026-09-14 21:19:06,085 INFO mapreduce.Job: The url to track the job: http://DESKTOP-P503RAM.localdomain:8088/proxy/application_1789397904369_0002/
2026-09-14 21:19:06,096 INFO mapreduce.Job: Running job: job_1789397904369_0002
2026-09-14 21:19:23,428 INFO mapreduce.Job: Job job_1789397904369_0002 running in uber mode : false
2026-09-14 21:19:23,429 INFO mapreduce.Job: map 0% reduce 0%
2026-09-14 21:19:52,019 INFO mapreduce.Job: map 80% reduce 0%
2026-09-14 21:20:16,136 INFO mapreduce.Job: map 100% reduce 0%
```

Result:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -ls /air_quality/output/MR2_HighestPollutionLocation
Found 2 items
-rw-r--r-- 1 aceracer supergroup 0 2026-09-14 21:20 /air_quality/output/MR2_HighestPollutionLocation/_SUCCESS
-rw-r--r-- 1 aceracer supergroup 3961260 2026-09-14 21:20 /air_quality/output/MR2_HighestPollutionLocation/part-00000
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ ls -lh mapreduce/MR2_HighestPollutionLocation/output
total 3.8M
-rwxrwxrwx 1 aceracer aceracer 0 Sep 14 21:22 _SUCCESS
-rwxrwxrwx 1 aceracer aceracer 3.8M Sep 14 21:22 part-00000
```

Hadoop Overview Datanodes Datanode Volume Failures Snapshot Startup Progress Utilities ▾

Browse Directory

Show 25 entries Search:

☐		Permission		Owner		Group		Size		Last Modified		Replication		Block Size		Name	
☐		-rw-r--r--		aceracer		supergroup		0 B		Sep 14 21:20		1		128 MB		_SUCCESS	
☐		-rw-r--r--		aceracer		supergroup		3.78 MB		Sep 14 21:20		1		128 MB		part-00000	

Showing 1 to 2 of 2 entries Previous 1 Next

Hadoop, 2022.

7.3 MR3 — Pollution Trend by Year

Execution:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hadoop jar /mnt/c/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.2.4.jar \
-input /air_quality/input/Air_Quality.csv \
-output /air_quality/output/MR3_PollutionTrend \
-mapper /mnt/d/Air_Quality_Big_Data_Analysis/mapreduce/MR3_PollutionTrend/mapper.py \
-reducer /mnt/d/Air_Quality_Big_Data_Analysis/mapreduce/MR3_PollutionTrend/reducer.py
packageJobJar: [/tmp/hadoop-unjar5421550352900134630/] [] /tmp/streamjob4330356255321325989.jar tmpDir=null
2026-09-14 21:27:06,306 INFO client.RMProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-09-14 21:27:07,313 INFO client.RMProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-09-14 21:27:07,979 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/aceracer/.staging/job_1789397904369_0003
2026-09-14 21:27:08,735 INFO mapred.FileInputFormat: Total input files to process : 1
2026-09-14 21:27:08,858 INFO mapreduce.JobSubmitter: number of splits:7
2026-09-14 21:27:09,066 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789397904369_0003
2026-09-14 21:27:09,074 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-14 21:27:09,727 INFO conf.Configuration: resource-types.xml not found
2026-09-14 21:27:09,730 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-14 21:27:10,017 INFO impl.YarnClientImpl: Submitted application application_1789397904369_0003
2026-09-14 21:27:10,262 INFO mapreduce.Job: The url to track the job: http://DESKTOP-P503RAM.localdomain:8088/proxy/application_1789397904369_0003/
2026-09-14 21:27:10,274 INFO mapreduce.Job: Running job: job_1789397904369_0003
2026-09-14 21:27:30,698 INFO mapreduce.Job: Job job_1789397904369_0003 running in uber mode : false
2026-09-14 21:27:30,700 INFO mapreduce.Job: map 0% reduce 0%
2026-09-14 21:28:06,710 INFO mapreduce.Job: map 12% reduce 0%
2026-09-14 21:28:07,986 INFO mapreduce.Job: map 26% reduce 0%
2026-09-14 21:28:11,107 INFO mapreduce.Job: map 52% reduce 0%
2026-09-14 21:28:14,322 INFO mapreduce.Job: map 79% reduce 0%
2026-09-14 21:28:15,340 INFO mapreduce.Job: map 86% reduce 0%
2026-09-14 21:28:35,564 INFO mapreduce.Job: map 100% reduce 0%
2026-09-14 21:28:39,625 INFO mapreduce.Job: map 100% reduce 100%
2026-09-14 21:28:40,645 INFO mapreduce.Job: Job job_1789397904369_0003 completed successfully
2026-09-14 21:28:41,233 INFO mapreduce.Job: Counters: 55
File System Counters
FILE: Number of bytes read=58886982
```

Result:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -cat /air_quality/output/MR3_PollutionTrend/part-00000 | head -30
2000|1,1,2,2-Tetrachloroethane 0.0129
2000|1,1,2-Trichloro-1,2,2-trifluoroethane 0.1472
2000|1,1,2-Trichloroethane 0.0079
2000|1,1-Dichloroethane 0.0070
2000|1,1-Dichloroethylene 0.0190
2000|1,2,3-Trimethylbenzene 0.8631
2000|1,2,4-Trichlorobenzene 0.1084
2000|1,2,4-Trimethylbenzene 2.1682
2000|1,2-Dichlorobenzene 0.1681
2000|1,2-Dichloroethene(total) 0.0001
2000|1,2-Dichloropropane 0.0336
2000|1,3,5-Trimethylbenzene 0.6274
2000|1,3-Butadiene 0.5376
2000|1,3-Dichlorobenzene 0.1043
2000|1,3-Dichloropropane(total) 0.0000
2000|1,3-dinitropyrene (TSP) STP 0.7465
2000|1,4-Dichlorobenzene 0.4854
2000|1-Butene 1.4563
2000|1-Decene 0.0000
2000|1-Heptene 0.4526
2000|1-Hexene 0.5476
2000|1-Hexene & 2-Methyl-1-pentene 0.0000
2000|1-Nonene 0.2541
2000|1-Octene 0.2114
2000|1-Pentene 0.6954
2000|1-Undecene 0.3098
2000|1-nitropyrene (TSP) STP 0.3788
2000|2,2,3-Trimethylbutane 0.5081
2000|2,2,4-Trimethylpentane 2.3676
2000|2,2-Dimethylbutane 0.6126
cat: Unable to write to output stream.
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -get /air_quality/output/MR3_PollutionTrend mapreduce/MR3_PollutionTrend/output
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ ls -lh mapreduce/MR3_PollutionTrend/output
total 352K
-rwxrwxrwx 1 aceracer aceracer 0 Sep 14 21:30 SUCCESS
-rwxrwxrwx 1 aceracer aceracer 350K Sep 14 21:30 part-00000
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$
```

Hadoop

Overview

Datanodes

Datanode Volume Failures

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/air_quality/output/MR3_PollutionTrend/part-00000

Go!

Show 25 entries

Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	-rw-r--r--	aceracer	supergroup	350 KB	Sep 14 21:28	1	128 MB		

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Hadoop, 2022.

8.Pig Version Verification:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ pig -version
Apache Pig version 0.17.0 (r1797386)
compiled Jun 02 2017, 15:41:58
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$
```

9. Part B · Apache Pig Analysis (5 jobs)

9.1 Pig 1 — Average Pollutant Concentration

Source Code:

```
GNU nano 8.7.1 pig/P1_AveragePollutant/p1_average_pollutant.pig
-- Pig Job 1: Average Pollutant Concentration

data = LOAD '/air_quality/pig_input/Air_Quality_Pig.csv'
      USING PigStorage(',')
      AS (
        Year:int,
        State:chararray,
        County:chararray,
        Parameter:chararray,
        Units:chararray,
        ArithmeticMean:double,
        MaxValue:double,
        Latitude:double,
        Longitude:double
      );

-- Remove header row
clean_data = FILTER data BY Year IS NOT NULL;

-- Group records by pollutant
grouped = GROUP clean_data BY Parameter;

-- Calculate average concentration
result = FOREACH grouped GENERATE
  group AS Parameter,
  AVG(clean_data.ArithmeticMean) AS Average_Concentration;

-- Sort by pollutant name
sorted_result = ORDER result BY Parameter ASC;

STORE sorted_result INTO '/air_quality/pig_output/P1_AveragePollutant'
  USING PigStorage('\t');
```

Execution:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ pig -f pig/P1_AveragePollutant/p1_average_pollutant.pig
2026-09-14 22:07:33,868 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-14 22:07:33,878 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-14 22:07:33,878 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-14 22:07:34,275 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-14 22:07:34,276 [main] INFO org.apache.pig.Main - Logging error messages to: /mnt/d/Air_Quality_Big_Data_Analysis/pig_1789402054240.log
2026-09-14 22:07:36,695 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file /home/aceracer/.pigbootstrap not found
2026-09-14 22:07:37,152 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.job.tracker.address
2026-09-14 22:07:37,153 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-14 22:07:43,295 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-p1_average_pollutant.pig-98027782-be77-4e49-861d-857099122b56
2026-09-14 22:07:43,297 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-14 22:07:47,453 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-14 22:07:47,688 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER
2026-09-14 22:07:47,865 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-14 22:07:48,204 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCaster, MergeFilter, MergeForEach, NestedJoinOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCaster]}
2026-09-14 22:07:48,603 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699408192 to monitor: collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-14 22:07:48,983 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.LMRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-14 22:07:49,228 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-14 22:07:49,344 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.LMR - Using Secondary Key Optimization for MapReduce node scope=50
2026-09-14 22:07:49,405 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.LMR - MR plan size before optimization: 3
2026-09-14 22:07:49,405 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.LMR - MR plan size after optimization: 3
2026-09-14 22:07:49,980 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at localhost:12270.0.0.1:8032
2026-09-14 22:07:52,416 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-14 22:07:52,517 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-14 22:07:52,570 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-14 22:07:52,584 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.LMR - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-14 22:07:52,599 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-14 22:07:52,606 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.LMR - Reduce phase detected, estimating # of required reducers.
2026-09-14 22:07:52,606 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.LMR - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapreduce.LMR - Inputs
2026-09-14 22:07:52,658 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.LMR - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=169502492
2026-09-14 22:07:52,658 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.LMR - JobControlManager - Getting ResultList for Job
```

Output:

```
2026-09-14 22:19:59,290 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.LMR - Success!
2026-09-14 22:19:59,478 [main] INFO org.apache.pig.Main - Pig script completed in 12 minutes, 25 seconds and 988 milliseconds (745988 ms)
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -cat /air_quality/pig_output/P1_AveragePollutant/part-* | head -30
1,1,1,2,2-Pentafluoroethane 0.601095795460126
1,1,1,2-Tetrachloroethane 0.007678827956989247
1,1,1-Trichloro-2,2-bis (p-chlorophenyl) ethane 6.397777777777777E-4
1,1,2,2-Tetrachloroethane 0.009018788027611606
1,1,2-Trichloro-1,2,2-trifluoroethane 0.15053736350677108
1,1,2-Trichloroethane 0.007434864714424039
1,1-Bis(4-chlorophenyl)-2,2-dichloroethane 6.397777777777777E-4
1,1-Dichloroethane 0.009074287257019451
1,1-Dichloroethenylidene bis(4-chlorobenzene) 0.04712633333333334
1,1-Dichloroethylene 0.009276896981312944
1,1-Difluoroethane 0.0546850674846623
1,2,3,4,5,6-Hexachlorocyclohexane,.beta.- 6.397777777777777E-4
1,2,3,4,5,6-Hexachlorocyclohexane 6.397777777777777E-4
1,2,3,4,5,6-Hexachlorocyclohexane,.delta.- 6.397777777777777E-4
1,2,3-Trichloropropane 3.7357142857142857E-4
1,2,3-Trimethylbenzene 0.4633641126257704
1,2,4,5-Tetrachlorobenzene (TSP) STD 0.0
1,2,4,5-Tetramethylbenzene 0.22912849999999998
1,2,4-Trichlorobenzene 0.0677612368090452
1,2,4-Trichlorobenzene (TSP) STD 0.0
1,2,4-Trimethylbenzene 0.9885477406996971
1,2-Butadiene 0.01630015
1,2-Dichlorobenzene 0.1300314074814309
1,2-Dichlorobenzene (TSP) STD 0.063986
1,2-Dichloroethene(total) 0.014277176470588236
1,2-Dichloropropane 0.02086851027319943
1,2-Epoxybutane 0.002429285714285714
1,3,5-Trichlorobenzene 9.34E-4
1,3,5-Trimethylbenzene 0.22776358427864055
1,3-Butadiene 0.2707358947637286
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ mkdir -p pig/P1_AveragePollutant/output
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -get /air_quality/pig_output/P1_AveragePollutant/part-r-00000 pig/P1_AveragePollutant/output/
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ ls -lh pig/P1_AveragePollutant/output
total 44K
-rwxrwxrwx 1 aceracer aceracer 42K Sep 14 22:24 part-r-00000
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$
```



9.2 Pig 2: Location Analysis

Source Code:

```
aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis
GNU nano 8.7.1 pig/P2_LocationAnalysis/p2_location...
-- Pig Job 2: Location Analysis

data = LOAD '/air_quality/pig_input/Air_Quality_Pig.tsv'
      USING PigStorage('\t')
      AS (
        Year:int,
        State:chararray,
        County:chararray,
        Parameter:chararray,
        Units:chararray,
        ArithmeticMean:double,
        MaxValue:double,
        Latitude:double,
        Longitude:double
      );

-- Remove header row
clean_data = FILTER data BY Year IS NOT NULL;

-- Group by State and County
grouped = GROUP clean_data BY (State, County);

-- Calculate average pollutant concentration for each location
result = FOREACH grouped GENERATE
  group.State AS State,
  group.County AS County,
  AVG(clean_data.ArithmeticMean) AS Average_Concentration;

-- Sort locations by State and County
sorted_result = ORDER result BY State ASC, County ASC;

STORE sorted_result INTO '/air_quality/pig_output/P2_LocationAnalysis'
USING PigStorage('\t');
```

Execution:

```
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ pig -f pig/P2_LocationAnalysis/p2_location_analysis.pig
2026-09-14 22:26:55,362 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-14 22:26:55,371 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-14 22:26:55,371 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-14 22:26:55,709 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-14 22:26:55,709 [main] INFO org.apache.pig.Main - Logging error messages to: /mnt/d/Air_Quality_Big_Data_Analysis/pig_1789403215678.log
2026-09-14 22:26:58,019 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file /home/aceracer/.pigbootstrap not found
2026-09-14 22:26:58,420 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-14 22:26:58,420 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-14 22:27:03,542 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-P2_LocationAnalysis.pig-83729c67-2d72-428f-aef8-bc35218cb306
2026-09-14 22:27:03,542 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-14 22:27:07,448 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-14 22:27:07,594 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP BY,ORDER BY,FILTER
2026-09-14 22:27:07,822 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-14 22:27:08,126 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-09-14 22:27:08,530 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-14 22:27:08,906 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-14 22:27:09,169 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-14 22:27:09,306 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope-56
2026-09-14 22:27:09,379 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MR plan size before optimization: 3
2026-09-14 22:27:09,379 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-14 22:27:09,966 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-09-14 22:27:12,101 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-14 22:27:12,210 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-14 22:27:12,264 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-14 22:27:12,264 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-14 22:27:12,291 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compressor is deprecated. Instead, use mapreduce.output.fileoutputformat.compressor
```

Output:

```
2026-09-14 22:39:34,587 [main] INFO org.apache.pig.Main - Pig script completed in 12 minutes, 39 seconds and 575 milliseconds (759575 ms)
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -cat /air_quality/pig_output/P2_LocationAnalysis/part-* | head -20
Alabama Baldwin 91.656088230315
Alabama Barbour 34.617141958955216
Alabama Clay 152.46629019398011
Alabama Colbert 110.02422130210762
Alabama DeKalb 82.5409871456834
Alabama Elmore 1.1909860222222224
Alabama Escambia 192.9253573125
Alabama Etowah 117.40523701999979
Alabama Houston 134.8476413217392
Alabama Jackson 12.321455862745099
Alabama Jefferson 75.24148776879248
Alabama Lawrence 4.332699140478669
Alabama Limestone 2.9821925185185183
Alabama Madison 86.01006241513285
Alabama Marengo 21.37355133333335
Alabama Mobile 61.8325772788901
Alabama Montgomery 84.63055874449655
Alabama Morgan 102.58831792134836
Alabama Pike 99.79903829126211
Alabama Russell 67.29249045965241
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ mkdir -p pig/P2_LocationAnalysis/output
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -get /air_quality/pig_output/P2_LocationAnalysis/part-r-00000 pig/P2_LocationAnalysis/output/
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ ls -lh pig/P2_LocationAnalysis/output
total 52K
-rwxrwxrwx 1 aceracer aceracer 51K Sep 14 22:41 part-r-00000
```

Download Head the file (first 32K) Tail the file (last 32K)

Block information -- Block 0

Block ID: 1073742358

Block Pool ID: BP-787223300-127.0.1.1-1789029273100

Generation Stamp: 1534

Size: 51822

Availability:

DESKTOP-P503RAM.localdomain

9.3 Pig 3: Pollution Trend by Year

Source Code:

```
aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis
GNU nano 8.7.1 pig/P3_PollutionTrend/p3_pollution_trend.pig
-- Pig Job 3: Pollution Trend

data = LOAD '/air_quality/pig_input/Air_Quality_Pig.tsv'
USING PigStorage('\t')
AS (
    Year:int,
    State:chararray,
    County:chararray,
    Parameter:chararray,
    Units:chararray,
    ArithmeticMean:double,
    MaxValue:double,
    Latitude:double,
    Longitude:double
);

-- Remove header row
clean_data = FILTER data BY Year IS NOT NULL;

-- Group by Year and Pollutant
grouped = GROUP clean_data BY (Year, Parameter);

-- Calculate average concentration
result = FOREACH grouped GENERATE
    group.Year AS Year,
    group.Parameter AS Parameter,
    AVG(clean_data.ArithmeticMean) AS Average_Concentration;

-- Sort by year and pollutant
sorted_result = ORDER result BY Year ASC, Parameter ASC;

STORE sorted_result INTO '/air_quality/pig_output/P3_PollutionTrend'
USING PigStorage('\t');
```

Execution:

```
(kaggle-env) aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis$ pig -f pig/P3_PollutionTrend/p3_pollution_trend.pig
2026-09-14 22:43:38,777 [main] INFO org.apache.pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-14 22:43:38,786 [main] INFO org.apache.pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-14 22:43:38,786 [main] INFO org.apache.pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-14 22:43:39,146 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1707386) compiled Jun 02 2017, 15:41:58
2026-09-14 22:43:39,146 [main] INFO org.apache.pig.Main - Logging error messages to: /mnt/d/Air_Quality_Big_Data_Analysis/pig-178940421912.log
2026-09-14 22:43:41,654 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file /home/aceracer/.pigbootstrap not found
2026-09-14 22:43:42,043 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-14 22:43:42,043 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-14 22:43:51,987 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-p3_pollution_trend.pig-64d65f54-3476-4433-b633-b3d6b3ed6243
2026-09-14 22:43:47,671 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-14 22:43:51,613 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-14 22:43:51,748 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER
2026-09-14 22:43:51,987 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig:schematuple] was not set... will not generate code.
2026-09-14 22:43:52,324 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - (RULES_ENABLED[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter])
2026-09-14 22:43:52,324 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699408192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-14 22:43:53,130 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCCompiler - File concatenation threshold: 100 optimistic? false
2026-09-14 22:43:53,384 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-14 22:43:53,522 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope-56
2026-09-14 22:43:53,593 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-09-14 22:43:53,593 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-14 22:43:54,185 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-09-14 22:43:56,423 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-14 22:43:56,423 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-14 22:43:56,571 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-14 22:43:56,571 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-14 22:43:56,586 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-14 22:43:56,605 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-14 22:43:56,614 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputS
```

Output:

```
2026-09-14 22:56:07,970 [main] INFO org.apache.pig.Main - Pig script completed in 12 minutes, 29 seconds and 536 milliseconds (749536 ms)
(kaggle-env) aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -cat /air_quality/pig_output/P3_PollutionTrend/part-* | head -30
2000 1,1,2,2-Tetrachloroethane 0.012909866666666676
2000 1,1,2-Trichloro-1,2,2-trifluoroethane 0.1471765
2000 1,1,2-Trichloroethane 0.00787938202247191
2000 1,1-Dichloroethane 0.006904731428571429
2000 1,1-Dichloroethylene 0.018994389830508475
2000 1,2,3-Trimethylbenzene 0.8631344734299511
2000 1,2,4-Trichlorobenzene 0.1083692307692308
2000 1,2,4-Trimethylbenzene 2.1681959406250013
2000 1,2-Dichlorobenzene 0.16810137162162161
2000 1,2-Dichloroethene (total) 0.00911225
2000 1,2-Dichloropropane 0.03359136112849161
2000 1,3,5-Trimethylbenzene 0.6274206750788646
2000 1,3-Butadiene 0.5375940723404259
2000 1,3-Dichlorobenzene 0.104330896
2000 1,3-Dichloropropene (total) 0.0
2000 1,3-dinitropyrene (TSP) STP 0.7454668
2000 1,4-Dichlorobenzene 0.48535808917197476
2000 1-Butene 1.4563329148936162
2000 1-Decene 0.0
2000 1-Heptene 0.45263256249999995
2000 1-Hexene 0.5476278536585366
2000 1-Hexene & 2-Methyl-1-pentene 0.0
2000 1-Nonene 0.25413025
2000 1-Octene 0.2114004375
2000 1-Pentene 0.6953860975609756
2000 1-Undecene 0.3097604375
2000 1-nitropyrene (TSP) STP 0.3788332
2000 2,2,3-Trimethylbutane 0.5080911538461538
2000 2,2,4-Trimethylpentane 2.367628819512195
2000 2,2-Dimethylbutane 0.6126389268292683
cat: Unable to write to output stream.
(kaggle-env) aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis$ mkdir -p pig/P3_PollutionTrend/output
(kaggle-env) aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -get /air_quality/pig_output/P3_PollutionTrend/part-r-00000 pig/P3_PollutionTrend/output/
(kaggle-env) aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis$ ls -ln pig/P3_PollutionTrend/output
total 456k
-rwxr-xr-x 1 aceracer aceracer 454K Sep 14 22:59 part-r-00000
(kaggle-env) aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis$ _
```

File information - part-r-00000

[Download](#)[Head the file \(first 32K\)](#)[Tail the file \(last 32K\)](#)

Block information — Block 0 ▾

Block ID: 1073742397

Block Pool ID: BP-787223300-127.0.1.1-1789029273100

Generation Stamp: 1573

Size: 463895

Availability:

- DESKTOP-P503RAM.localdomain

9.4 Pig 4: Top 10 Polluted Locations

Source Code:

```
Select aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis
GNU nano 8.7.1 pig/P4_TopPollutedLocations/p4
-- Pig Job 4: Top 10 Polluted Locations

data = LOAD '/air_quality/pig_input/Air_Quality_Pig.tsv'
      USING PigStorage('\t')
      AS (
        Year:int,
        State:chararray,
        County:chararray,
        Parameter:chararray,
        Units:chararray,
        ArithmeticMean:double,
        MaxValue:double,
        Latitude:double,
        Longitude:double
      );

-- Remove header row
clean_data = FILTER data BY Year IS NOT NULL AND MaxValue IS NOT NULL;

-- Group by pollutant and location
grouped = GROUP clean_data BY (Parameter, State, County);

-- Find the highest pollution value for each pollutant-location combination
result = FOREACH grouped GENERATE
  group.Parameter AS Parameter,
  group.State AS State,
  group.County AS County,
  MAX(clean_data.MaxValue) AS Highest_Value;

-- Sort from highest to lowest
sorted_result = ORDER result BY Highest_Value DESC;

-- Select top 10
top10 = LIMIT sorted_result 10;

STORE top10 INTO '/air_quality/pig_output/P4_TopPollutedLocations'
  USING PigStorage('\t');
```

Execution:

```
(kaggle-env) aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis$ nano pig/P4_TopPollutedLocations/p4_top_polluted_locations.pig
(kaggle-env) aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis$ pig -f pig/P4_TopPollutedLocations/p4_top_polluted_locations.pig
2026-09-14 23:01:19,411 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-14 23:01:19,421 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-14 23:01:19,421 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-14 23:01:19,802 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-14 23:01:19,802 [main] INFO org.apache.pig.Main - Logging error messages to: /mnt/d/Air_Quality_Big_Data_Analysis/pig_1789405279768.log
2026-09-14 23:01:22,392 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file /home/aceracer/.pigbootstrap not found
2026-09-14 23:01:22,858 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-14 23:01:22,858 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-14 23:01:29,256 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-p4_top_polluted_locations.pig-20b5073e-6d11-40ec-8d3a-0a905fd44b88
2026-09-14 23:01:33,668 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-14 23:01:33,804 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY, ORDER_BY, FILTER, LIMIT
2026-09-14 23:01:34,066 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-14 23:01:34,431 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - (RULES_ENABLED=AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCaster, MergeFilter, MergeForEach, NestedJoinOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushMapFilter, SplitFilter, StreamTypeCaster)
2026-09-14 23:01:34,853 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-14 23:01:35,271 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-14 23:01:35,520 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-14 23:01:35,664 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope-63
2026-09-14 23:01:35,731 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-14 23:01:35,732 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-14 23:01:36,289 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-09-14 23:01:38,050 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-14 23:01:38,732 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-14 23:01:38,801 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-14 23:01:38,802 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-14 23:01:38,815 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-14 23:01:38,830 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-14 23:01:38,837 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-14 23:01:38,884 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=169502492
2026-09-14 23:01:38,884 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-14 23:01:38,884 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reducers
```

Output:

```
2026-09-14 23:18:05,180 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Unable to retrieve job to compute warning aggregation.
2026-09-14 23:18:05,180 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-14 23:18:05,365 [main] INFO org.apache.pig.Main - Pig script completed in 16 minutes, 46 seconds and 328 milliseconds (1006328 ms)
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -cat /air_quality/pig_output/P4_TopPollutedLocations/part-
Particle Number, Total Count Rhode Island Providence 336583.0
Particle Number, Total Count New York Queens 194576.0
Particle Number, Total Count Maryland Howard 185521.0
Particle Number, Total Count New York Erie 185345.0
Particle Number, 30-50 nm Florida Broward 171798.0
Particle Number, Total Count New York Albany 168226.0
Particle Number, 20-30 nm Florida Broward 142048.0
Particle Number, Total Count New York Monroe 139595.0
Particle Number, Total Count Florida Hillsborough 109709.0
Particle Number, 50-70 nm Florida Broward 100353.0
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ mkdir -p pig/P4_TopPollutedLocations/output
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -get /air_quality/pig_output/P4_TopPollutedLocations/part-r-00000 pig/P4_TopPollutedLocations/output/
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ ls -lh pig/P4_TopPollutedLocations/output/
total 0
-rwxr-xr-x 1 aceracer aceracer 542 Sep 14 23:19 part-r-00000
```

File information - part-r-00000

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Block information —

Block 0 ▾

Block ID: 1073742449

Block Pool ID: BP-787223300-127.0.1.1-1789029273100

Generation Stamp: 1625

Size: 542

Availability:

- DESKTOP-P503RAM.localdomain

9.5 Pig 5: Pollutant Analysis

Source Code:

```
aceracer@DESKTOP-P503RAM: /mnt/d/Air_Quality_Big_Data_Analysis
GNU nano 8.7.1 pig/P5_PollutantAnalysis/p5
-- Pig Job 5: Pollutant Analysis

data = LOAD '/air_quality/pig_input/Air_Quality_Pig.tsv'
      USING PigStorage('\t')
      AS (
        Year:int,
        State:chararray,
        County:chararray,
        Parameter:chararray,
        Units:chararray,
        ArithmeticMean:double,
        MaxValue:double,
        Latitude:double,
        Longitude:double
      );

-- Remove header row
clean_data = FILTER data BY Year IS NOT NULL;

-- Group by pollutant and measurement unit
grouped = GROUP clean_data BY (Parameter, Units);

-- Calculate observation count and average concentration
result = FOREACH grouped GENERATE
  group.Parameter AS Parameter,
  group.Units AS Units,
  COUNT(clean_data) AS Observation_Count,
  AVG(clean_data.ArithmeticMean) AS Average_Concentration;

-- Sort by pollutant name
sorted_result = ORDER result BY Parameter ASC;

STORE sorted_result INTO '/air_quality/pig_output/P5_PollutantAnalysis'
USING PigStorage('\t');
```

Execution:

```
rm: /air_quality/pig_output/P5_PollutantAnalysis/: No such file or directory
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ pig -f pig/P5_PollutantAnalysis/p5_pollutant_analysis.pig
2026-09-14 23:22:30.406 [main] INFO org.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-14 23:22:30.413 [main] INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-14 23:22:30.413 [main] INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-14 23:22:30.780 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-14 23:22:30.780 [main] INFO org.apache.pig.Main - Logging error messages to: /mnt/d/Air_Quality_Big_Data_Analysis/pig_1789406550748.log
2026-09-14 23:22:33.001 [main] INFO org.apache.pig.impl.util.Utilis - Default bootstrap file /home/aceracer/.pigbootstrap not found
2026-09-14 23:22:33.345 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.job.tracker.address
2026-09-14 23:22:33.347 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-14 23:22:38.598 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-p5_pollutant_analysis.pig-8ce2e18a-fceb-4d51-8662-15ecbc0ee687
2026-09-14 23:22:38.598 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-14 23:22:42.556 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-14 23:22:42.695 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER
2026-09-14 23:22:42.930 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-14 23:22:43.245 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushDownFilter, SplitFilter, StreamTypeCastInserter]}
2026-09-14 23:22:43.627 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699408192 to monitor. collection.sage.threshold = 489580128, usage.threshold = 489580128
2026-09-14 23:22:44.322 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-14 23:22:44.474 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope-58
2026-09-14 23:22:44.553 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-09-14 23:22:44.553 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-14 23:22:45.159 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-09-14 23:22:48.109 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-14 23:22:48.216 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-14 23:22:48.269 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-14 23:22:48.269 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-14 23:22:48.284 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-14 23:22:48.300 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
```

Output:

```
2026-09-14 23:34:54.293 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-14 23:34:54.437 [main] INFO org.apache.pig.Main - Pig script completed in 12 minutes, 24 seconds and 415 milliseconds (744415 ms)
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -cat /air_quality/pig_output/P5_PollutantAnalysis/part-* | head -30
1,1,1,2-Pentafluoroethane Parts per billion Carbon 93 0.0010957975460124
1,1,1,2-Tetrachloroethane Parts per billion Carbon 93 0.007678827956989248
1,1,1-Trichloro-2,2-bis (p-chlorophenyl) ethane Nanograms/cubic meter (25 C) 9 6.397777777777777E-4
1,1,2,2-Tetrachloroethane Parts per billion Carbon 4346 0.009018788027611634
1,1,2-Trichloro-1,2,2-trifluoroethane Parts per billion Carbon 1403 0.1505373635067711
1,1,2-Trichloroethane Parts per billion Carbon 4132 0.007434864714424041
1,1-Bis(4-chlorophenyl)-2,2-dichloroethane Nanograms/cubic meter (25 C) 9 6.397777777777777E-4
1,1-Dichloroethane Parts per billion Carbon 4167 0.009074287257019436
1,1-Dichloroethylenedibis(4-chlorobenzene) Nanograms/cubic meter (25 C) 9 0.04712633333333333
1,1-Dichloroethylene Parts per billion Carbon 4174 0.009276896981312927
1,1-Difluoroethane Parts per billion Carbon 163 0.8546850674846624
1,2,3,4,5,6-Hexachlorocyclohexane, .beta.- Nanograms/cubic meter (25 C) 9 6.397777777777777E-4
1,2,3,4,5,6-Hexachlorocyclohexane, .delta.- Nanograms/cubic meter (25 C) 9 6.397777777777777E-4
1,2,3-Trichloropropane Parts per billion Carbon 7 3.7357142857142857E-4
1,2,3-Trimethylbenzene Parts per billion Carbon 3081 0.4633641126257707
1,2,4,5-Tetrachlorobenzene (TSP) STP Nanograms/cubic meter (25 C) 13 0.0
1,2,4,5-Tetramethylbenzene Parts per billion Carbon 8 0.2291285
1,2,4-Trichlorobenzene Parts per billion Carbon 3184 0.06776123680904532
1,2,4-Trichlorobenzene (TSP) STP Nanograms/cubic meter (25 C) 13 0.0
1,2,4-Trimethylbenzene Parts per billion Carbon 6317 0.988547740699698
1,2-Butadiene Parts per billion Carbon 20 0.016300150000000003
1,2-Dichlorobenzene Parts per billion Carbon 3502 0.1300314074814391
1,2-Dichlorobenzene (TSP) STP Nanograms/cubic meter (25 C) 13 0.063986
1,2-Dichloroethene(total) Parts per billion Carbon 17 0.014277176470588234
1,2-Dichloropropane Parts per billion Carbon 4429 0.020868519273199463
1,2-Epoxybutane Parts per billion Carbon 7 0.002429285714285714
1,3,5-Trichlorobenzene Parts per billion Carbon 7 9.34E-4
1,3,5-Trimethylbenzene Parts per billion Carbon 6259 0.3277635842786391
1,3-Butadiene Parts per billion Carbon 5863 0.27073589476377263
cat: Unable to write to output stream
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ mkdir -p pig/P5_PollutantAnalysis/output
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ hdfs dfs -get /air_quality/pig_output/P5_PollutantAnalysis/part-r-00000 pig/P5_PollutantAnalysis/output/
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$ ls -lh pig/P5_PollutantAnalysis/output
total 76K
-rwxr-xrwx 1 aceracer aceracer 73K Sep 14 23:36 part-r-00000
(kaggle-env) aceracer@DESKTOP-P503RAM:/mnt/d/Air_Quality_Big_Data_Analysis$
```

File information - part-r-00000

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Block information -- Block 0

Block ID: 1073742488

Block Pool ID: BP-787223300-127.0.1.1-1789029273100

Generation Stamp: 1664

Size: 74004

Availability:

- DESKTOP-P503RAM.localdomain

10. showing successful completed applications:

Node Labels Applications NEW NEW_SAVING SUBMITTED ACCEPTED RUNNING FINISHED FAILED KILLED Scheduler Tools	Cluster Nodes Metrics													
	Active Nodes			Decommissioning Nodes			Decommissioned Nodes			Lost Nodes				
	1	0			0			0			0			
	Scheduler Metrics													
	Scheduler Type		Scheduling Resource Type				Minimum Allocation			Maximum Allocation				
	Capacity Scheduler		[memory-mb (unit=Mi), vcores]				<memory:1024, vCores:1>			<memory:8192, vCores:4>				
	Show 20 ▾ entries													
	ID	User	Name	Application Type	Queue	Application Priority	StartTime	LaunchTime	FinishTime	State	FinalStatus	Runn	Contain	
	application_1789397904369_0022	aceracer	PigLatin:p5_pollutant_analysis.pig	MAPREDUCE	default	0	Mon Sep 14 23:29:04 +0600 2026	Mon Sep 14 23:29:05 +0600 2026	Mon Sep 14 23:30:14 +0600 2026	FINISHED	SUCCEEDED	N/A		
	application_1789397904369_0021	aceracer	PigLatin:p5_pollutant_analysis.pig	MAPREDUCE	default	0	Mon Sep 14 23:26:12 +0600 2026	Mon Sep 14 23:26:12 +0600 2026	Mon Sep 14 23:27:28 +0600 2026	FINISHED	SUCCEEDED	N/A		
application_1789397904369_0020	aceracer	PigLatin:p5_pollutant_analysis.pig	MAPREDUCE	default	0	Mon Sep 14 23:22:54 +0600 2026	Mon Sep 14 23:22:54 +0600 2026	Mon Sep 14 23:24:36 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0019	aceracer	PigLatin:p4_top_polluted_locations.pig	MAPREDUCE	default	0	Mon Sep 14 23:11:04 +0600 2026	Mon Sep 14 23:11:04 +0600 2026	Mon Sep 14 23:12:28 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0018	aceracer	PigLatin:p4_top_polluted_locations.pig	MAPREDUCE	default	0	Mon Sep 14 23:08:02 +0600 2026	Mon Sep 14 23:08:02 +0600 2026	Mon Sep 14 23:09:14 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0017	aceracer	PigLatin:p4_top_polluted_locations.pig	MAPREDUCE	default	0	Mon Sep 14 23:04:59 +0600 2026	Mon Sep 14 23:04:59 +0600 2026	Mon Sep 14 23:06:25 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0016	aceracer	PigLatin:p4_top_polluted_locations.pig	MAPREDUCE	default	0	Mon Sep 14 23:01:45 +0600 2026	Mon Sep 14 23:01:46 +0600 2026	Mon Sep 14 23:03:25 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0015	aceracer	PigLatin:p3_pollution_trend.pig	MAPREDUCE	default	0	Mon Sep 14 22:50:08 +0600 2026	Mon Sep 14 22:50:09 +0600 2026	Mon Sep 14 22:51:30 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0014	aceracer	PigLatin:p3_pollution_trend.pig	MAPREDUCE	default	0	Mon Sep 14 22:47:10 +0600 2026	Mon Sep 14 22:47:11 +0600 2026	Mon Sep 14 22:48:34 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0013	aceracer	PigLatin:p3_pollution_trend.pig	MAPREDUCE	default	0	Mon Sep 14 22:44:02 +0600 2026	Mon Sep 14 22:44:03 +0600 2026	Mon Sep 14 22:45:31 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0012	aceracer	PigLatin:p2_location_analysis.pig	MAPREDUCE	default	0	Mon Sep 14 22:33:29 +0600 2026	Mon Sep 14 22:33:29 +0600 2026	Mon Sep 14 22:34:54 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0011	aceracer	PigLatin:p2_location_analysis.pig	MAPREDUCE	default	0	Mon Sep 14 22:30:26 +0600 2026	Mon Sep 14 22:30:26 +0600 2026	Mon Sep 14 22:31:50 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0010	aceracer	PigLatin:p2_location_analysis.pig	MAPREDUCE	default	0	Mon Sep 14 22:27:18 +0600 2026	Mon Sep 14 22:27:18 +0600 2026	Mon Sep 14 22:28:49 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0009	aceracer	PigLatin:p1_average_pollutant.pig	MAPREDUCE	default	0	Mon Sep 14 22:13:59 +0600 2026	Mon Sep 14 22:14:00 +0600 2026	Mon Sep 14 22:15:21 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0008	aceracer	PigLatin:p1_average_pollutant.pig	MAPREDUCE	default	0	Mon Sep 14 22:11:01 +0600 2026	Mon Sep 14 22:11:02 +0600 2026	Mon Sep 14 22:12:21 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0007	aceracer	PigLatin:p1_average_pollutant.pig	MAPREDUCE	default	0	Mon Sep 14 22:07:58 +0600 2026	Mon Sep 14 22:07:59 +0600 2026	Mon Sep 14 22:09:27 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0006	aceracer	PigLatin:p1_average_pollutant.pig	MAPREDUCE	default	0	Mon Sep 14 21:50:53 +0600 2026	Mon Sep 14 21:50:54 +0600 2026	Mon Sep 14 21:51:36 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0005	aceracer	PigLatin:p1_average_pollutant.pig	MAPREDUCE	default	0	Mon Sep 14 21:48:36 +0600 2026	Mon Sep 14 21:48:36 +0600 2026	Mon Sep 14 21:49:16 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0004	aceracer	PigLatin:p1_average_pollutant.pig	MAPREDUCE	default	0	Mon Sep 14 21:45:58 +0600 2026	Mon Sep 14 21:45:58 +0600 2026	Mon Sep 14 21:46:58 +0600 2026	FINISHED	SUCCEEDED	N/A			
application_1789397904369_0003	aceracer	streamiob4330356255321325989.1ar	MAPREDUCE	default	0	Mon Sep 14 21:43:00 +0600 2026	Mon Sep 14 21:43:00 +0600 2026	Mon Sep 14 21:44:00 +0600 2026	FINISHED	SUCCEEDED	N/A			

Conclusion

This project successfully demonstrated the use of Big Data technologies for analyzing a large-scale air-quality dataset using the Hadoop ecosystem. Approximately 1.69 million air-quality records collected over 19 years were prepared and stored in the Hadoop Distributed File System (HDFS).

Hadoop MapReduce was used to perform three major analyses. The first MapReduce job calculated average pollutant concentrations, the second identified locations with high recorded pollution values, and the third analyzed pollution trends by year. These jobs demonstrated how Hadoop can process large datasets efficiently using distributed processing.

Apache Pig was also used to perform five additional analyses, including average pollutant concentration, location analysis, pollution trends, top polluted locations, and pollutant analysis. The successful execution of these Pig tasks demonstrated the use of Pig Latin for high-level processing of large datasets.

The Hadoop services, HDFS, MapReduce jobs, Pig jobs, and YARN applications were verified through the command line and Hadoop localhost interfaces. The successful completion of the applications confirms that the Big Data environment was properly configured and the implemented jobs were executed successfully.

This project provided practical experience in storing, processing, and analyzing large environmental datasets using Hadoop, HDFS, MapReduce, Apache Pig, and YARN. The project also demonstrated how Big Data technologies can be applied to environmental data to identify pollutant patterns, analyze locations, and study changes in air quality over time.